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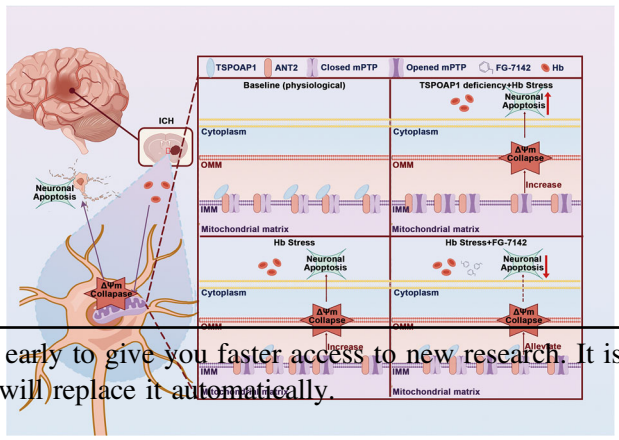
Loss of TSPOAP1 exacerbates mitochondrial dysfunction and neuronal apoptosis after intracerebral hemorrhage

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Abstract

Intracerebral hemorrhage (ICH) induces severe secondary neuronal injury, in which mitochondrial dysfunction and apoptosis play central roles. Accumulating evidence suggests that mitochondrial failure represents a critical convergence point linking diverse injurious stimuli to neuronal apoptosis after ICH. However, the molecular mechanisms that connect mitochondrial dysfunction to irreversible apoptotic commitment remain incompletely understood. In particular, the upstream regulators governing mitochondrial permeability transition pore (mPTP) opening during ICH have yet to be fully defined. Transcriptomic analyses of perihematomal tissue from patients with ICH and mouse models were integrated with functional studies using an HT22 neuronal cell line deficient in 18-kDa translocator protein (TSPO)-associated protein 1 (TSPOAP1), together with pharmacological intervention using FG-7142. Mitochondrial integrity, mPTP opening, apoptotic signaling, and protein–protein interactions were evaluated in vitro and in vivo. The mitochondria-associated protein TSPOAP1 was markedly downregulated after ICH in both human and mouse brains and was predominantly expressed in neurons. TSPOAP1 deficiency was associated with substantial mitochondrial impairment, evidenced by membrane depolarization, ultrastructural abnormalities, elevated mPTP opening, and increased neuronal apoptosis, along with disruption of the Bax/Bcl-2 balance and cytochrome c leakage. Loss of TSPOAP1 sensitized mPTP opening and was accompanied by increased mitochondrial oxidative stress, while remaining relatively insensitive to cyclophilin D (CypD) inhibition. Mechanistically, TSPOAP1 was associated with adenine nucleotide translocator 2 (ANT2). FG-7142 increased TSPOAP1 expression, maintained mitochondrial homeostasis, suppressed neuronal apoptosis in vitro, and promoted functional recovery after ICH in vivo. Notably, these beneficial effects were markedly attenuated in the absence of TSPOAP1. This study identifies TSPOAP1 as a previously unrecognized mitochondrial protective factor that influences neuronal apoptosis after ICH by regulating ANT2-associated, CypD-independent mPTP opening. Enhancing TSPOAP1 function represents a promising therapeutic strategy for mitigating mitochondrial dysfunction and secondary brain injury following ICH.



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Introduction

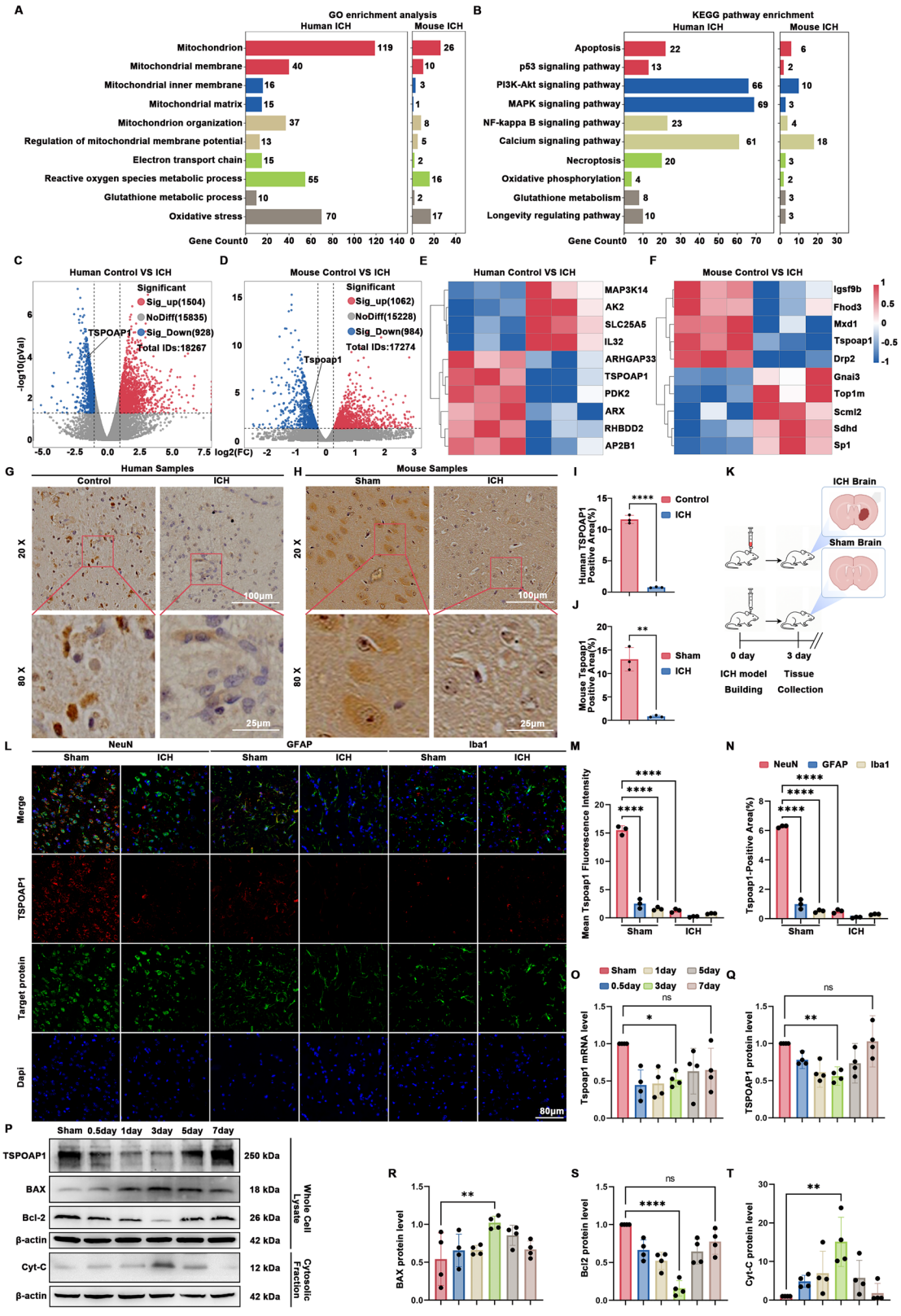
ICH accounts for approximately 15–20% of all strokes and is associated with a particularly poor prognosis, with mortality rates reaching up to 50% [1–3]. Beyond the primary mechanical injury caused by hematoma formation, secondary brain damage—driven by excitotoxicity, oxidative stress, neuroinflammation, and mitochondrial dysfunction—plays a pivotal role in determining neurological outcomes [4–6]. Among these pathological processes, neuronal apoptosis has emerged as a central contributor to perihematomal tissue loss [7–9]. Accumulating evidence indicates that mitochondrial injury, particularly aberrant opening of the mPTP, represents a critical initiating event in ICH-induced neuronal death [10–12]. However, the upstream molecular determinants governing mPTP sensitivity in the context of ICH remain incompletely understood [13, 14].

Traditional models of mPTP control have centered on excessive ROS generation and CypD engagement as principal drivers of mitochondrial depolarization, cytochrome c efflux, and caspase activation [15–18]. However, the molecular identity of the pore remains incompletely resolved. Growing evidence supports a role for the mitochondrial F_1F_0-ATP synthase in pore formation, with proposed models involving ATP synthase dimers, monomers, or the c-subunit ring, whereas ANT family members may also form or regulate permeability pathways under specific conditions [10, 19–22]. These observations support a more heterogeneous model of permeability transition in which multiple inner-membrane components can determine pore susceptibility depending on cellular and metabolic context. In particular, ANT isoforms can critically influence pore susceptibility without strict reliance on CypD, supporting the existence of alternative regulatory routes governing mPTP opening [23–27]. Such mechanisms may be particularly relevant in ICH, where strategies exclusively targeting ROS or CypD have yielded limited therapeutic benefit [28, 29].

TSPOAP1 was originally identified as an interacting partner of the 18-kDa translocator protein (TSPO) [30, 31]. However, TSPO and TSPOAP1 exhibit distinct cellular distributions within the central nervous system. Consistent with previous reports, TSPO expression is largely confined to activated microglia [31, 32]. However, our data demonstrate that TSPOAP1 is predominantly expressed in neurons, indicating a TSPO-independent role in the brain. Although TSPOAP1 has been well characterized for its roles in presynaptic organization and neuronal excitability [33–36], its potential involvement in mitochondrial homeostasis remains largely undefined. Recent observations have pointed to a possible link between TSPOAP1 and the regulation of mitochondrial membrane permeability [37, 38], raising the possibility that TSPOAP1 influences neuronal survival by modulating susceptibility to mitochondrial permeability transition through mechanisms that remain to be clarified.

FG-7142 is a well-characterized partial inverse agonist at the benzodiazepine allosteric site of $GABA_A$ receptors and has been extensively used in experimental models of affective and stress-related disorders [39, 40]. Importantly, FG-7142 readily crosses the blood–brain barrier following systemic administration and exerts robust central nervous system effects [41–43], supporting its suitability for pharmacological evaluation in brain injury models. In our experiments, FG-7142 was predicted to interact with TSPOAP1 and was found to increase TSPOAP1 protein expression. These observations prompted us to investigate whether FG-7142 could serve as a pharmacological tool to modulate TSPOAP1 and thereby influence mitochondrial injury and neurological outcomes after ICH.

In this study, we identify TSPOAP1 as a previously unrecognized mitochondrial protective factor in ICH. Through integrated analyses of human perihematomal tissue, mouse ICH models, and a TSPOAP1-deficient neuronal cell line, we demonstrate that loss of TSPOAP1 is associated with enhanced neuronal apoptosis and mitochondrial dysfunction following hemorrhagic injury. Our findings suggest that TSPOAP1 influences mPTP susceptibility through an ANT2-associated and CypD-insensitive pathway. Furthermore, we show that pharmacological upregulation of TSPOAP1 using FG-7142 confers robust mitochondrial and neuroprotective effects both in vitro and in vivo. Collectively, our



study highlights TSPOAP1 as a key regulator of neuronal fate and FG-7142 as a potential therapeutic agent for alleviating neuronal apoptosis after ICH.

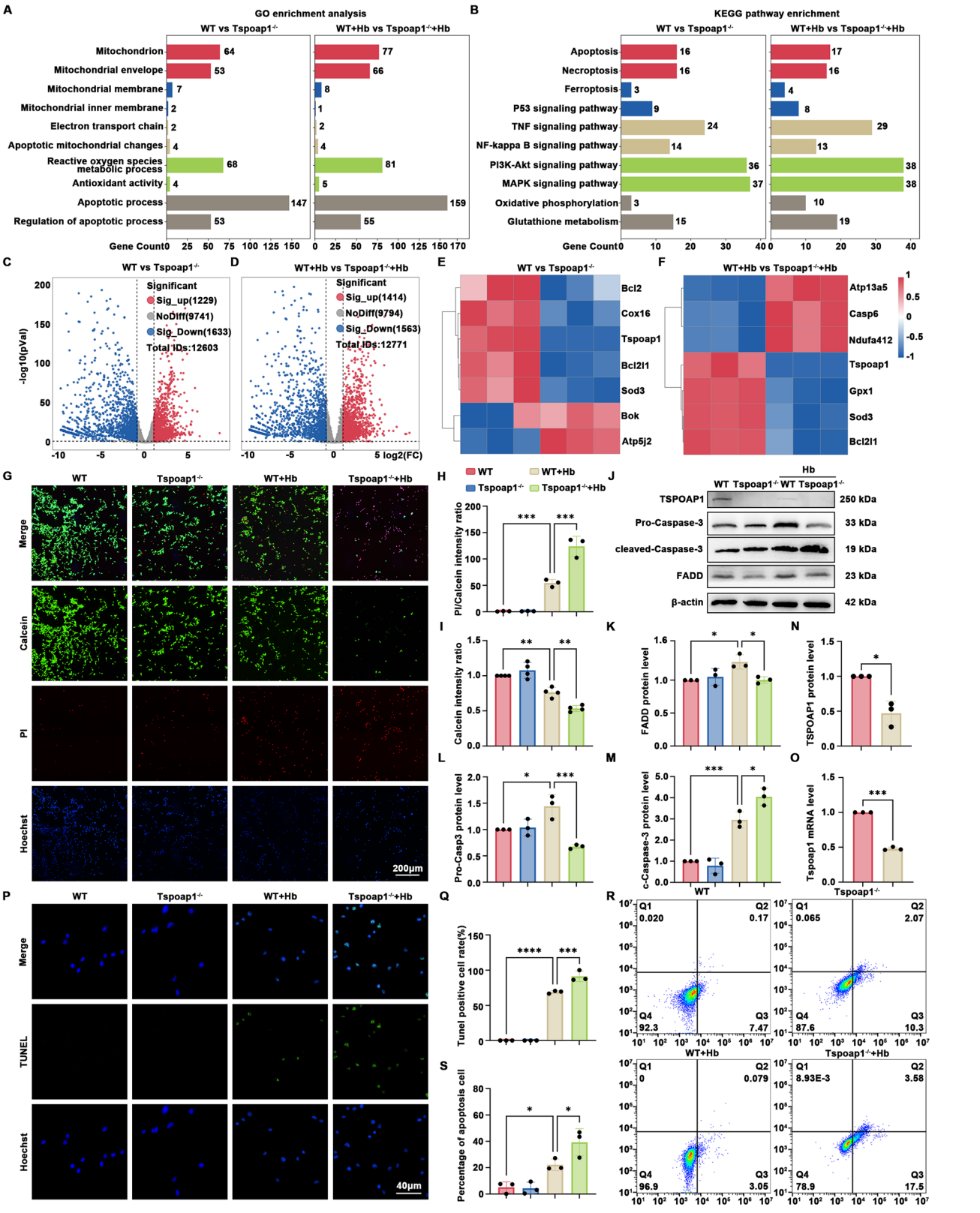
◀**Fig. 1 TSPOAP1 is downregulated in human and mouse perihematomal brain tissue after intracerebral hemorrhage.** **A**, **B** Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analyses of differentially expressed genes from human and mouse ICH transcriptomic datasets, highlighting mitochondrial-, oxidative stress-, and cell death-related biological processes and signaling pathways. Numbers indicate the corresponding gene counts. Volcano plots showing differentially expressed genes between control and ICH samples in the human (**C**) and mouse (**D**) datasets. Red and blue dots indicate significantly upregulated and downregulated genes, respectively. TSPOAP1/Tspoap1 is highlighted. Heatmaps showing the expression patterns of representative mitochondria- and apoptosis-associated genes in human (**E**) and mouse (**F**) control and ICH samples. Representative immunohistochemical staining of TSPOAP1 in human (**G**) and mouse (**H**) brain tissues under control or sham and ICH conditions. Upper panels show low-magnification images (×20), and lower panels show enlarged views (×80). Scale bars, 100 μm and 25 μm, respectively. Quantification of TSPOAP1-positive area in human (**I**) and mouse (**J**) brain tissues, $n = 3$. **K** Schematic illustration of the experimental timeline for establishment of the mouse ICH model and tissue collection. **L** Representative immunofluorescence images showing TSPOAP1 (red) co-stained with NeuN, GFAP, or Iba1 (green) in sham and ICH mouse brain sections. Nuclei were stained with DAPI (blue). Scale bar, 80 μm. Quantification of mean TSPOAP1 fluorescence intensity (**M**) and TSPOAP1-positive area (**N**) in NeuN-positive neurons, GFAP-positive astrocytes, and Iba1-positive microglia, $n = 3$. **O** Quantitative analysis of Tspoap1 mRNA expression in mouse perihematomal brain tissue at the indicated time points after ICH, $n = 4$. **P** Representative immunoblots showing TSPOAP1, BAX, and Bcl-2 expression in whole-cell lysates and cytochrome c expression in the cytosolic fraction of mouse perihematomal brain tissue at the indicated time points after ICH. β-actin was used as a loading control. Quantification of TSPOAP1 **Q**, BAX **R**, Bcl-2 **S**, and cytosolic cytochrome c (**T**) protein levels, $n = 4$. Data are presented as mean ± SD. Statistical significance is indicated as follows: $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, and $****P < 0.0001$; ns not significant.

Results

TSPOAP1 is downregulated in perihematomal neurons after intracerebral hemorrhage

Comparative transcriptomic profiling of perilesional brain tissues from human ICH patients and a parallel murine model delineated significant alterations in gene expression profiles (Fig. 1A–F). Functional annotation across both species revealed that the predominant genomic shifts involved mitochondrial structural integrity, mitochondrial membrane potential($\Delta\Psi m$) maintenance, and redox homeostasis, with several functional clusters converging on mitochondrial-driven cell death (Fig. 1A). Furthermore, KEGG enrichment identified a high prevalence of signaling cascades including apoptosis, p53, and PI3K–AKT (Fig. 1B). These findings collectively highlight a robust induction of mitochondrial-dependent apoptotic networks within the perihematomal microenvironment. Detailed pathway information is provided in the Supplementary Materials (Fig. S1A, B). Cross-species comparison identified a subset of consistently dysregulated genes in both human and mouse ICH datasets (Fig. 1C, D), among which TSPOAP1 was one of the most prominently downregulated. Heatmap analysis of representative mitochondrial- and apoptosis-related genes revealed highly concordant expression patterns between species (Fig. 1E, F), supporting conserved mitochondrial apoptotic responses after ICH.

To confirm the transcriptomic results at the protein level, we performed IHC in perihematomal tissue from patients with ICH and the corresponding mouse model. TSPOAP1 immunoreactivity was markedly decreased in perihematomal regions compared with controls in both species (Fig. 1G–J). IF co-staining revealed that TSPOAP1 was predominantly localized to NeuN-positive neurons, with minimal overlap with GFAP-positive astrocytes or Iba1-positive microglia (Fig. 1L). Following ICH, the reduction of TSPOAP1 immunoreactivity was predominantly observed in NeuN-positive neurons (Fig. 1M, N). Time-course analysis in mouse perihematomal tissue showed that Tspoap1 mRNA levels declined during the early post-ICH phase and gradually recovered thereafter (Fig. 1O). Western blotting demonstrated a similar pattern at the protein level, with decreased TSPOAP1 expression at 1–3 days after ICH and partial restoration by day 7 (Fig. 1P, Q). Importantly, Bcl-2 expression reached its lowest level at day 3 after ICH, whereas Bax expression peaked at the same time point, coinciding with the nadir of TSPOAP1 protein expression (Fig. 1P, R, S). In addition, increased cytosolic cytochrome c levels further indicated activation of mitochondrial apoptotic signaling after ICH (Fig. 1P, T). These findings support a temporal association between early TSPOAP1 downregulation and activation of mitochondrial apoptotic pathways following ICH.


A
GO enrichment analysis
WT vs Tspoap1-/-
WT+Hb vs Tspoap1-/-+Hb
Mitochondrion 64 77
Mitochondrial envelope 53 66
Mitochondrial membrane 7 8
Mitochondrial inner membrane 2 1
Electron transport chain 2 2
Apoptotic mitochondrial changes 4 4
Reactive oxygen species metabolic process 68 81
Antioxidant activity 4 5
Apoptotic process 147 159
Regulation of apoptotic process 53 55
Gene Count
B
KEGG pathway enrichment
WT vs Tspoap1-/-
WT+Hb vs Tspoap1-/-+Hb
Apoptosis 16 17
Necroptosis 16 16
Ferroptosis 3 4
P53 signaling pathway 9 8
TNF signaling pathway 24 29
NF-kappa B signaling pathway 14 13
PI3K-Akt signaling pathway 36 38
MAPK signaling pathway 37 38
Oxidative phosphorylation 3 10
Glutathione metabolism 15 19
Gene Count
C
WT vs Tspoap1-/-
Significant
Sig_up(1229)
NoDiff(9741)
Sig_Down(1633)
Total IDs:12603
-log10(pVal)
D
WT+Hb vs Tspoap1-/-+Hb
Significant
Sig_up(1414)
NoDiff(9794)
Sig_Down(1563)
Total IDs:12771
5 log2(FC)
E
WT vs Tspoap1-/-
Bcl2
Cox16
Tspoap1
Bcl2l1
Sod3
Bok
Atp5j2
F
WT+Hb vs Tspoap1-/-+Hb
Atp13a5
Casp6
Ndufa412
Tspoap1
Gpx1
Sod3
Bcl2l1
G
WT
Tspoap1-/-
WT+Hb
Tspoap1-/-+Hb
Merge
Calcein
PI
Hoechst
200μm
H
WT
WT+Hb
Tspoap1-/-
Tspoap1-/-+Hb
PI/Calcein intensity ratio
J
Hb
WT Tspoap1-/- WT Tspoap1-/-
TSPOAP1 250 kDa
Pro-Caspase-3 33 kDa
cleaved-Caspase-3 19 kDa
FADD 23 kDa
β-actin 42 kDa
I
Calcein intensity ratio
K
FADD protein level
N
TSPOAP1 protein level
L
Pro-Casp3 protein level
M
c-Caspase-3 protein level
O
Tspoap1 mRNA level
P
WT
Tspoap1-/-
WT+Hb
Tspoap1-/-+Hb
Merge
TUNEL
Hoechst
40μm
Q
Tunel positive cell rate(%)
R
WT
Tspoap1-/-
WT+Hb
Tspoap1-/-+Hb
Q1 0.020
Q2 0.17
Q4 92.3
Q3 7.47
Q1 0.065
Q2 2.07
Q4 87.6
Q3 10.3
Q1 0
Q2 0.079
Q4 96.9
Q3 3.05
Q1 8.93E-3
Q2 3.58
Q4 78.9
Q3 17.5
S
Percentage of apoptosis cell

◀**Fig. 2 TSPOAP1 deficiency exacerbates Hb-induced neuronal injury and apoptosis in vitro. A** GO enrichment analysis of DEGs across WT and *Tspoap1*$^{-/-}$ cohorts under baseline or Hb-treated conditions, highlighting mitochondrial, oxidative stress, and apoptotic processes. **B** KEGG pathway enrichment analysis of DEGs between WT and *Tspoap1*$^{-/-}$ cells, including ferroptosis, necroptosis, and stress-related signaling pathways. Volcano plots depicting DEGs between WT and *Tspoap1*$^{-/-}$ cells under basal (**C**) or Hb-treated (**D**) conditions. Heatmaps showing expression patterns of selected mitochondrial and apoptosis-related genes under control (**E**) and Hb-treated (**F**) conditions. **G** Representative images of Calcein-AM/PI co-staining in WT and *Tspoap1*$^{-/-}$ HT22 cells. Nuclei were counterstained with Hoechst. Scale bar, 200 μm. **H** Quantification of PI/Calcein fluorescence intensity ratios derived from imaging data, $n = 3$. **I** Microplate-based quantification of intracellular calcein fluorescence, $n = 4$. **J** Representative immunoblot images showing TSPOAP1, FADD, and pro/cleaved caspase-3 expression. β-actin serves as a loading control. **K**–**M** Densitometric analysis of protein levels shown in panel (**J**). Quantification of TSPOAP1 protein (**N**) and mRNA (**O**) expression, $n = 3$. Quantitative comparisons were limited to WT groups due to negligible TSPOAP1 signals in knockout cells. **P** Representative confocal images of TUNEL staining. Nuclei were counterstained with Hoechst. Scale bar, 40 μm. **Q** Quantification of TUNEL-positive cells. **R** Representative Annexin V/PI flow cytometry plots, $n = 3$. **S** Quantification of total apoptotic cell populations, $n = 3$. Data are presented as mean ± SD. Statistical significance is indicated as described above.

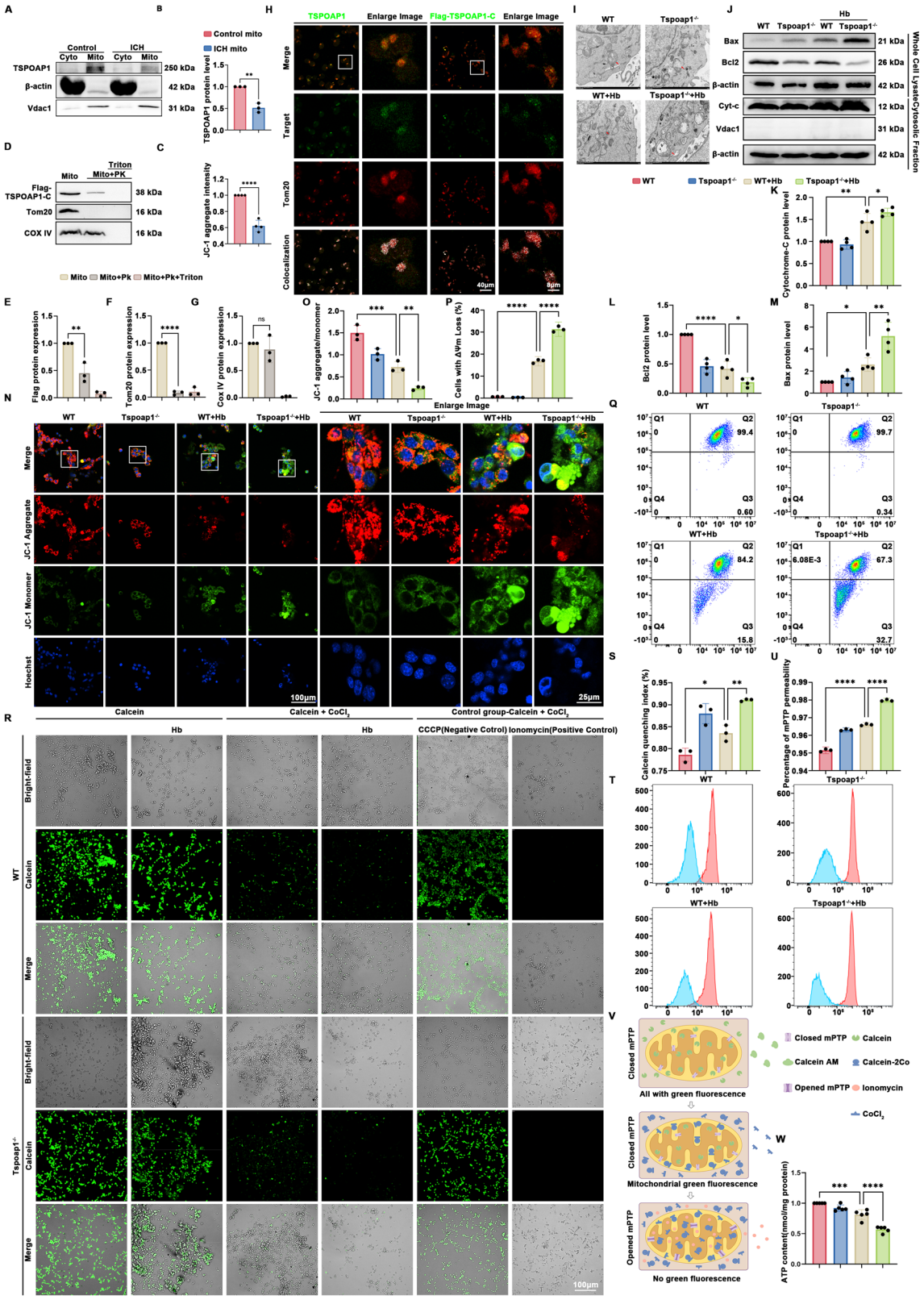
Loss of TSPOAP1 exacerbates hemoglobin-induced neuronal injury and apoptosis in vitro

To examine the impact of TSPOAP1 loss in neuronal injury, we generated *Tspoap1*$^{-/-}$ HT22 cells using CRISPR/Cas9. Transcriptomic profiling revealed broad gene expression alterations in knockout cells, which became more pronounced following Hb exposure (Fig. 2A–F). Functional annotation of the Tspoap1$^{-/-}$ transcriptome revealed a predominant clustering of altered genes around mitochondrial structural organization, redox homeostasis, and the control of membrane potential and apoptotic pathways. These biological themes closely mirror the genomic shifts identified in clinical and experimental ICH specimens. Furthermore, pathway enrichment via KEGG delineated a robust activation of multiple cell death subroutines—including ferroptosis, necroptosis, and apoptosis—as well as calcium-related signaling, particularly when cells were subjected to Hb challenge (Fig. 2B). Principal component analysis further substantiated a clear transcriptomic divergence between genotypes under both baseline and stimulated states (Fig. 2C, D). Collectively, the concordant expression patterns of mitochondrial and apoptotic markers between knockout cells and peri-hematomal tissues (Fig. 2E, F) suggest that TSPOAP1 deficiency recapitulates the core molecular landscape of hemorrhagic brain injury.

To assess cellular injury following TSPOAP1 loss, Calcein-AM/PI staining was performed in WT and *Tspoap1*$^{-/-}$ HT22 cells with or without Hb exposure (Fig. 2G). A preliminary dose–response assay identified 40 μmol/L Hb as an appropriate injury condition (Fig. S2). Upon Hb treatment, *Tspoap1*$^{-/-}$ cells displayed increased PI uptake together with reduced intracellular Calcein fluorescence compared with WT cells, indicating enhanced membrane damage (Fig. 2G). Quantitative analysis showed a significant increase in the PI/Calcein fluorescence ratio in TSPOAP1-deficient cells (Fig. 2H), which was further supported by a reduction in Calcein signal measured by microplate assay (Fig. 2I). TSPOAP1 depletion was confirmed at both the protein and mRNA levels under basal and Hb-treated conditions (Fig. 2J, N, O). Consistent with increased apoptotic signaling, *Tspoap1*$^{-/-}$ cells exhibited reduced pro-caspase-3 levels and elevated cleaved caspase-3 following Hb exposure (Fig. 2J, L, M). In contrast, expression of the adaptor protein FADD was decreased in TSPOAP1-deficient cells (Fig. 2J, K). At the cellular level, TUNEL staining revealed a higher proportion of apoptotic cells in *Tspoap1*$^{-/-}$ cultures, particularly after Hb stimulation (Fig. 2P, Q). This increase in apoptosis was further confirmed by Annexin V/PI flow cytometric analysis under both basal and Hb-treated conditions (Fig. 2R, S).

TSPOAP1 deficiency disrupts mitochondrial integrity and promotes mitochondrial apoptosis

To examine the subcellular distribution of TSPOAP1, cytosolic and mitochondrial fractions were isolated and subjected to immunoblot analysis. TSPOAP1 was predominantly detected in the mitochondrial fraction, whereas only low levels were observed in the cytosolic fraction (Fig. 3A, B). We next assessed $\Delta\Psi$m in vivo using mitochondria isolated from control and ICH mouse brain tissue. Microplate-based JC-1 analysis revealed a significant reduction in



JC-1 aggregate fluorescence intensity in ICH samples compared with controls, indicating dissipation of ΔΨm following hemorrhagic injury (Fig. 3C). Considering that previous studies have suggested that the TSPOAP1

◀**Fig. 3 TSPOAP1 is mitochondria-associated, and its loss aggravates mitochondrial dysfunction, permeability transition, and apoptotic signaling. A** Representative immunoblots showing TSPOAP1 in cytosolic (Cyto) and mitochondrial (Mito) fractions isolated from control and ICH mouse brain tissues. β-actin and VDAC1 were used as cytosolic and mitochondrial fraction markers, respectively. **B** Quantification of TSPOAP1 protein abundance in the mitochondrial fraction from control and ICH mouse brain tissues, $n = 3$. **C** Microplate-based JC-1 analysis of mitochondria isolated from control and ICH mouse brain tissues, showing JC-1 aggregate fluorescence intensity, $n = 4$. **D** Proteinase K protection assay of the FLAG-tagged mouse TSPOAP1 C-terminal fragment spanning amino acids 1561–1846 (FLAG-TSPOAP1-C) in isolated mitochondria. TOM20 and COX IV were used as outer mitochondrial membrane and inner mitochondrial membrane controls, respectively. Triton X-100 was used to permeabilize mitochondrial membranes. Quantification of FLAG-TSPOAP1-C (**E**) TOM20 (**F**) and COX IV (**G**) protein signals following the indicated treatments, $n = 3$. **H** Representative immunofluorescence images showing the spatial relationship of endogenous TSPOAP1 or FLAG-TSPOAP1-C with TOM20-positive mitochondrial structures. Enlarged views of the boxed regions are shown. Scale bars, 40 μm and 8 μm, respectively. **I** Representative transmission electron microscopy images of mitochondria in WT and Tspoap1$^{-/-}$ HT22 cells under basal and Hb-treated conditions. Red arrowheads indicate swollen mitochondria and disrupted cristae. Scale bar, 1 μm. **J** Representative immunoblots showing Bax and Bcl-2 expression in whole-cell lysates and cytochrome c expression in the cytosolic fraction of WT and Tspoap1$^{-/-}$ cells under basal and Hb-treated conditions. β-actin and VDAC1 were used as loading and fractionation controls, respectively. Quantification of cytosolic cytochrome c (**K**) Bcl-2 (**L**) and Bax (**M**) protein levels, $n = 4$. **N** Representative JC-1 fluorescence images showing mitochondrial membrane potential in WT and Tspoap1$^{-/-}$ cells under basal and Hb-treated conditions. Red indicates JC-1 aggregates, green indicates JC-1 monomers, and nuclei were stained with Hoechst. Enlarged views of the boxed regions are shown. Scale bars, 100 μm and 25 μm, respectively. **O** Quantification of the JC-1 aggregate-to-monomer fluorescence ratio, $n = 3$. **P** Quantification of cells exhibiting loss of mitochondrial membrane potential, $n = 3$. **Q** Representative flow cytometric plots of JC-1 staining in WT and *Tspoap1*$^{-/-}$ cells under basal and Hb-treated conditions. **R** Representative fluorescence images of the Calcein–$CoCl_2$ assay in WT and Tspoap1$^{-/-}$ cells. CCCP and ionomycin were used as negative and positive controls for mitochondrial calcein retention, respectively. Scale bar, 100 μm. **S** Quantification of the calcein fluorescence quenching index, $n = 3$. **T** Representative flow cytometric histograms showing mitochondrial calcein fluorescence distributions. **U** Quantification of cells exhibiting increased mPTP permeability, $n = 3$. **V** Schematic illustration of the Calcein–$CoCl_2$ assay principle under closed and opened mPTP conditions. **W** Cellular ATP content normalized to total protein in WT and Tspoap1$^{-/-}$ cells under basal and Hb-treated conditions, $n = 5$. Data are presented as mean ± SD. Statistical significance is indicated as described above.

C-terminal domain associates with the mitochondrial outer membrane protein TSPO [30], we designed and generated a FLAG-tagged mouse TSPOAP1 C-terminal construct spanning amino acids 1561–1846 (FLAG-TSPOAP1-C). Under conditions in which the outer mitochondrial membrane marker TOM20 was efficiently degraded by proteinase K, a fraction of FLAG-TSPOAP1-C remained detectable, whereas COX IV was largely protected. This result suggests that the TSPOAP1 C-terminal fragment may not only localize to the outer mitochondrial membrane but also localize to the mitochondrial matrix (Fig. 3D–H). In addition, immunofluorescence colocalization with TOM20 further supported the mitochondrial association of endogenous TSPOAP1 and FLAG-TSPOAP1-C (Fig. 3H), with quantitative Pearson correlation coefficients provided in Fig. S2C, D. These findings support TSPOAP1 localization to the mitochondria in HT22 cells.

Compared with WT cells, *Tspoap1*$^{-/-}$ HT22 cells exhibited pronounced mitochondrial structural abnormalities, including mitochondrial swelling and disrupted cristae organization, which were further exacerbated following Hb stimulation (Fig. 3I). These observations indicate that loss of TSPOAP1 compromises mitochondrial structural integrity. To further characterize the molecular phenotype associated with TSPOAP1 deficiency, we examined apoptosis-related mitochondrial proteins. WB result showed that loss of TSPOAP1 increased Bax/Bcl-2 expression level following Hb treatment, accompanied by enhanced accumulation of cytosolic cytochrome c (Fig. 3J–M), consistent with activation of a mitochondria-associated apoptotic program. Consistent with the aforementioned findings, JC-1 fluorescence profiling delineated a pronounced transition from mitochondrial aggregates to monomers within TSPOAP1- depleted cells post-Hb challenge (Fig. 3N). This shift from red to green fluorescence substantiates the widespread dissipation of $\Delta\Psi m$, further linking TSPOAP1 deficiency to organelle-level failure under hemorrhagic stress. Quantitative analysis of JC-1 imaging confirmed a significant decrease in the aggregate/monomer fluorescence ratio (Fig. 3O). Complementary flow cytometric analysis of JC-1 staining further demonstrated an increased proportion of cells exhibiting $\Delta\Psi m$ loss in *Tspoap1*$^{-/-}$ cells irrespective of Hb exposure (Fig. 3P, Q), supporting exacerbated mitochondrial depolarization upon TSPOAP1 loss. Finally, mPTP opening was directly assessed using the Calcein–$CoCl_2$ quenching assay. Representative fluorescence images showed markedly reduced

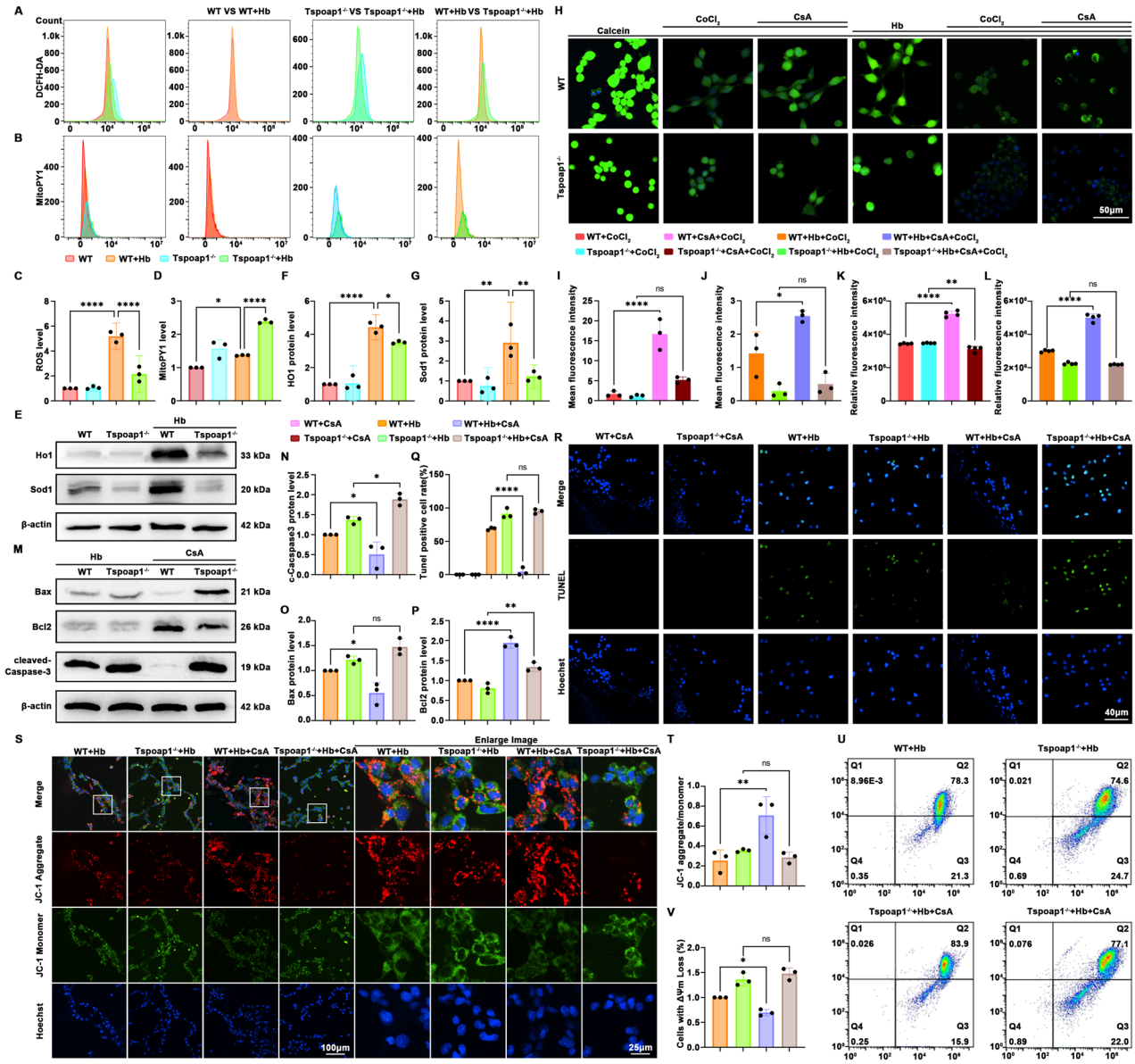


Fig. 4 TSPOAP1 loss enhances mitochondrial oxidative stress and promotes CypD-insensitive mPTP opening. Representative flow cytometry histograms showing intracellular ROS levels detected by DCFH-DA (**A**) and mitochondrial H_2O_2 detected by MitoPY1 (**B**). Quantification of intracellular ROS (**C**) and mitochondrial H_2O_2 levels (**D**). **E** Representative immunoblot images showing HO-1 and SOD1 expression in WT and Tspoap1$^{-/-}$ cells following Hb stimulation, $n = 3$. Quantification of HO-1 (**F**) and SOD1 (**G**) protein levels, $n = 3$. **H** Representative calcein fluorescence images under the indicated treatments. Scale bar, 50 μm. **I**, **J** Image-based quantification of mean calcein fluorescence intensity, $n = 3$. **K**, **L** Microplate-based quantification of calcein fluorescence intensity. **M** Representative immunoblot images showing Bax, Bcl-2, and cleaved caspase-3 expression under Hb and/or CsA treatment, $n = 4$. Quantification of cleaved caspase-3 (**N**) Bax (**O**) and Bcl-2 (**P**) levels. **Q** Quantification of TUNEL-positive cells, $n = 3$. **R** Representative TUNEL staining images. Scale bar, 40 μm. **S** Representative JC-1 fluorescence images. Scale bars as indicated. **T** Quantification of JC-1 aggregate/monomer fluorescence ratio. **U** Representative JC-1 flow cytometry plots, $n = 3$. **V** Quantification of cells exhibiting ΔΨm loss, $n = 3$. Data are presented as mean ± SD. Statistical significance is indicated as described above.

mitochondrial calcein retention in *Tspoap1*$^{-/-}$ cells compared with WT controls under both basal conditions and following Hb stimulation (Fig. 3R), indicating enhanced mPTP opening. Quantification of calcein fluorescence quenching demonstrated a significantly increased mPTP opening index in TSPOAP1-deficient cells (Fig. 3S). In parallel, flow cytometric analysis revealed a leftward shift in mitochondrial calcein fluorescence distributions

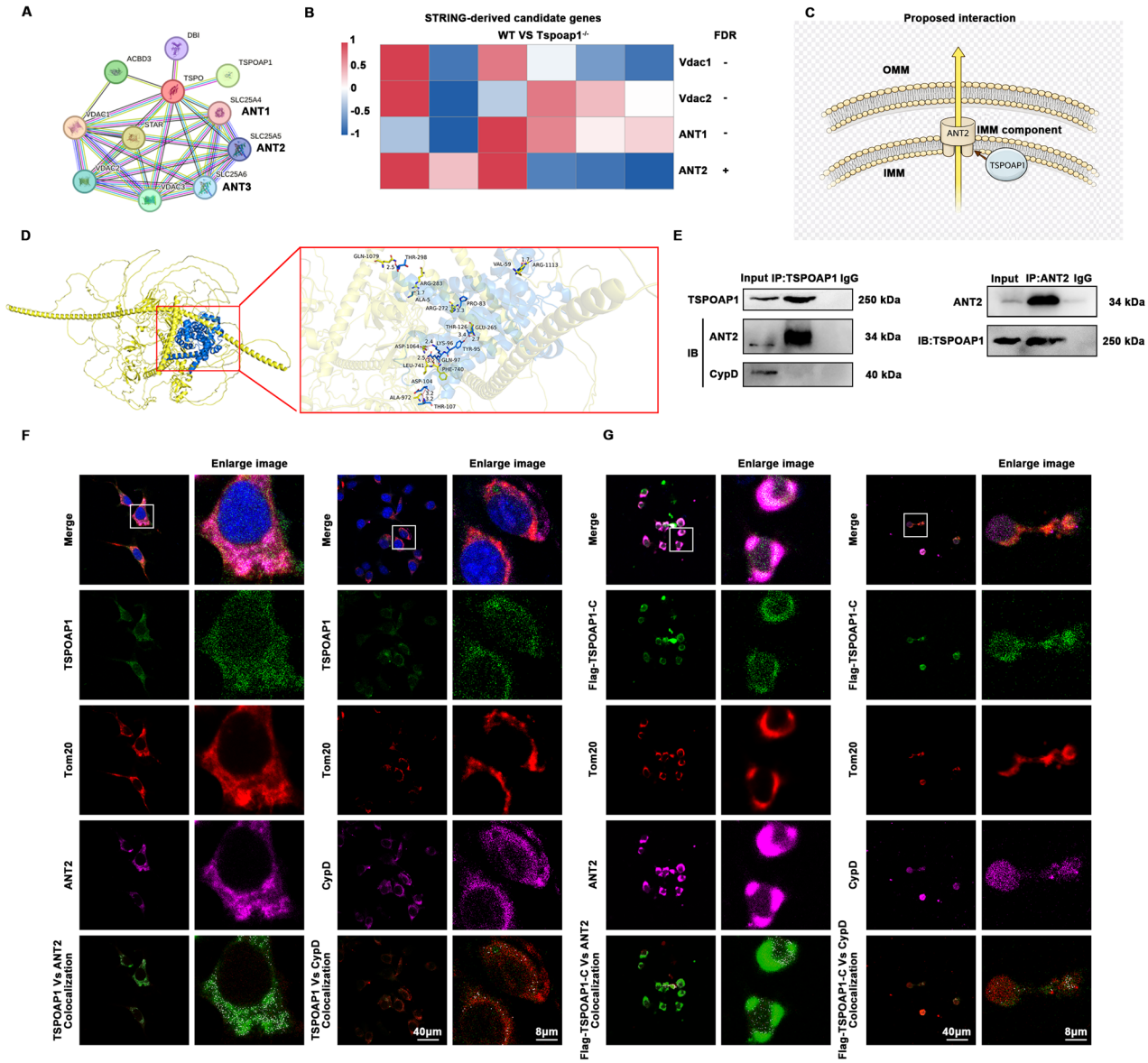


Fig. 5 TSPOAP1 associates with ANT2 in a mitochondria. **A** PPI network of TSPOAP1 generated using the STRING database, illustrating candidate TSPOAP1-associated proteins, including mitochondria- and membrane-related components. **B** Heatmap showing the relative expression of selected mitochondria-associated TSPOAP1-interacting candidates derived from the STRING network in WT versus *Tspoap1*$^{-/-}$ transcriptomic datasets. Genes meeting the differential expression threshold (FDR < 0.05) are indicated by "+", whereas those not meeting the threshold are indicated by "−". **C** Schematic illustration of the proposed association between TSPOAP1 and ANT2 in a mitochondria-associated context. OMM, outer mitochondrial membrane; IMM, inner mitochondrial membrane. The schematic represents a working model and does not establish the precise submitochondrial localization of TSPOAP1. **D** In silico protein–protein docking model of TSPOAP1 and ANT2 generated using the GRAMM platform. The top-ranked predicted conformation is shown, with a magnified view highlighting predicted contact residues at the putative interaction interface. The calculated binding free energy was −21.6 kcal/mol. **E** Co-IP analysis demonstrating the association of TSPOAP1 with ANT2, but not with CypD, in HT22 cells. Lysates were immunoprecipitated using antibodies against TSPOAP1, or ANT2, followed by immunoblotting with the indicated antibodies. **F** Representative immunofluorescence images showing the spatial distributions of endogenous TSPOAP1 relative to ANT2 and CypD in HT22 cells. Endogenous TSPOAP1 showed a stronger spatial correlation with ANT2 than with CypD, with representative Pearson correlation coefficients of 0.81 and 0.19, respectively. Scale bars: 40 μm (left panels) and 8 μm (right panels). **G** Representative immunofluorescence images showing the spatial distributions of FLAG-TSPOAP1-C relative to ANT2 and CypD in N2A cells. Representative Pearson correlation coefficients were 0.48 for FLAG-TSPOAP1-C–ANT2, and 0.37 for FLAG-TSPOAP1-C–CypD (Figure. S3C–F). Nuclei were counterstained with DAPI. Enlarged views of the boxed regions are shown. Scale bars: 40 μm (left panels) and 8 μm (right panels). The imaging and biochemical analyses support a preferential association of TSPOAP1 with ANT2 but do not establish direct binding or precise submitochondrial localization.

(Fig. 3T), and quantification confirmed a higher proportion of cells exhibiting increased mPTP permeability in the absence of TSPOAP1 (Fig. 3U). The underlying logic of the Calcein–$CoCl_2$ detection method is depicted in (Fig. 3V). In addition, cellular ATP content was reduced following Hb exposure and was further decreased in TSPOAP1-deficient cells, indicating aggravated mitochondrial energetic dysfunction under hemorrhagic stress (Fig. 3W). Taken together, these findings demonstrate that TSPOAP1 deficiency triggers mitochondrial dysfunction under Hb exposure conditions, characterized by ultrastructural damage, loss of membrane potential, increased permeability-transition susceptibility, suppressed ATP production, and subsequent activation of mitochondria-mediated apoptotic signaling.

TSPOAP1 loss enhances mitochondrial oxidative stress and promotes CypD-insensitive mPTP opening

To elucidate the regulatory role of TSPOAP1 in mPTP susceptibility, we initially examined whether the observed pore sensitization in TSPOAP1-deficient neurons was driven by fluctuations in oxidative stress. General intracellular oxidative signals and mitochondrial H_2O_2 accumulation were assessed using DCFH-DA and MitoPY1, respectively (Fig. 4A, B). Hb stimulation markedly increased DCFH-DA fluorescence in WT cells, whereas the corresponding signal was lower in Tspoap1$^{-/-}$ cells under Hb challenge (Fig. 4A, C). In contrast, MitoPY1 fluorescence was increased after TSPOAP1 loss and was further elevated following Hb exposure, with the highest mitochondrial H_2O_2 signal observed in Tspoap1$^{-/-}$ cells treated with Hb (Fig. 4B, D). These divergent probe responses suggest compartment-specific redox alterations rather than a uniform change in cellular ROS. While Hb stimulation increased HO-1 and SOD1 expression in WT cells, these antioxidant responses were attenuated in TSPOAP1-deficient cells (Fig. 4E–G). Together, these findings indicate that TSPOAP1 loss is associated with increased mitochondrial H_2O_2 accumulation and impaired antioxidant defense.

We next examined the sensitivity of mPTP opening to cyclosporin A (CsA). In WT cells, CsA markedly preserved mitochondrial calcein fluorescence, whereas Tspoap1$^{-/-}$ cells exhibited persistent calcein quenching under both basal and Hb-treated conditions (Fig. 4H–L). Consistently, CsA was less effective in attenuating Hb-induced Bax upregulation, caspase-3 cleavage, or Bcl-2 reduction in TSPOAP1-deficient cells (Fig. 4M–P), and did not reduce apoptotic as assessed by TUNEL staining (Fig. 4Q, R). Moreover, JC-1 imaging and flow cytometry demonstrated sustained mitochondrial depolarization in Tspoap1$^{-/-}$ cells despite CsA treatment (Fig. 4S–V).

Collectively, these data indicate that TSPOAP1 deficiency is accompanied by increased mitochondrial oxidative stress, impaired antioxidant responses, and enhanced permeability-transition susceptibility. The limited protective effect of CsA further supports the presence of a CypD-insensitive component of mPTP regulation after TSPOAP1 loss.

TSPOAP1 associates with ANT2 in mitochondria

To delineate the protein interactome of TSPOAP1, we initially mapped its putative associated network leveraging the STRING database (Fig. 5A). This computational profiling identified a set of candidate interaction partners, including several mitochondria-associated and membrane-associated proteins. A more exhaustive visualization of this TSPOAP1-centered interaction landscape is provided in Fig. S3A. To narrow down this pool to functional candidates most relevant to mitochondrial homeostatic control, we cross-referenced the STRING-derived interactors against our differential transcriptomic signatures from WT and *Tspoap1*$^{-/-}$ models. By specifically prioritizing mitochondria-associated transcripts (Fig. 5B), we were able to narrow the candidate list. Among these candidates, only ANT2 (Slc25a5) exhibited significant differential expression in the WT versus *Tspoap1*$^{-/-}$ comparison (FDR < 0.05), whereas other related components, including ANT1, and VDAC isoforms, did not meet the differential expression threshold. These results highlight ANT2 as a candidate TSPOAP1-associated mitochondrial protein.

Based on these observations, we proposed a working hypothesis that TSPOAP1 may associate with ANT2 in a mitochondrial context (Fig. 5C). To assess the structural feasibility of this interaction, we performed in silico protein–protein docking analysis using the GRAMM platform. The predicted TSPOAP1–ANT2 complex adopted a

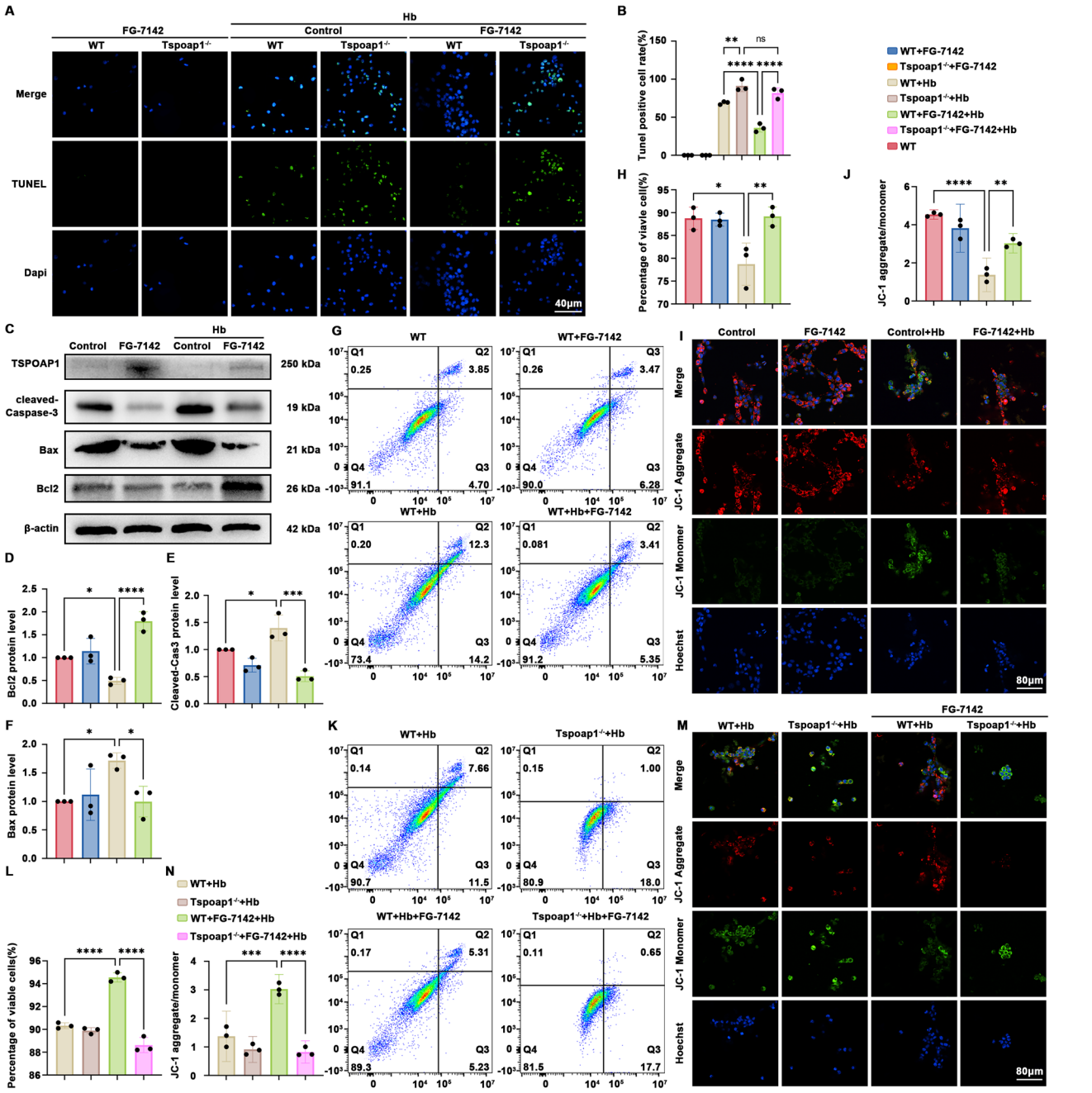


Fig. 6 FG-7142 attenuates mitochondrial dysfunction and apoptotic injury in a TSPOAP1-associated manner. **A** Representative TUNEL staining images of HT22 cells under the indicated treatments. Scale bar, 40 μm. **B** Quantification of TUNEL-positive cells. **C** Representative immunoblot images showing TSPOAP1, cleaved caspase-3, Bax, and Bcl-2 expression. **D–F** Quantification of protein levels shown in panel (**C**) $n = 3$. **G** Representative Annexin V/PI flow cytometry plots in WT cells. **H** Quantification of viable cell populations. **I** Representative JC-1 fluorescence images. Scale bar, 80 μm. **J** Quantification of JC-1 aggregate/monomer fluorescence ratios. **K** Representative Annexin V/PI flow cytometry plots in WT and Tspoap1$^{-/-}$ cells. **L** Quantification of viable cells across genotypes. **M** Representative JC-1 fluorescence images in WT and Tspoap1$^{-/-}$ cells treated with Hb and FG-7142. **N** Quantification of JC-1 aggregate/monomer fluorescence ratios. Data are presented as mean ± SD. Statistical significance is indicated as described above.

stable predicted binding conformation with a favorable predicted binding free energy −21.6 kcal/mol and a well-defined interaction interface (Fig. 5D; Fig. S3B). Notably, the identified contact residues were predominantly located within surface-accessible regions of ANT2, supporting the structural feasibility of TSPOAP1–ANT2 association.

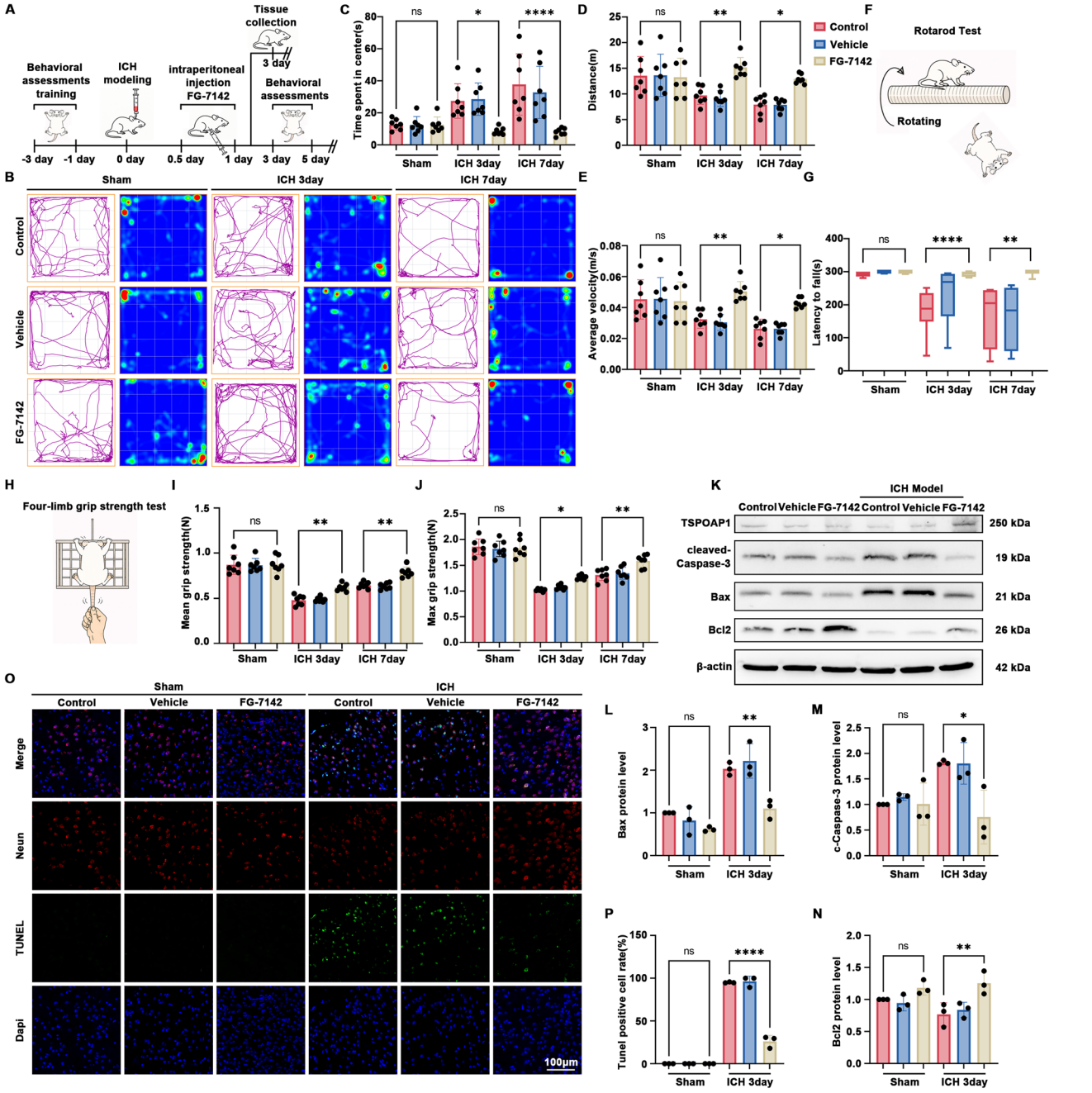


Fig. 7 FG-7142 improves neurological performance and reduces apoptosis injury after experimental intracerebral hemorrhage. A Schematic timeline illustrating the experimental design of the in vivo ICH model and FG-7142 treatment regimen. **B** Representative open field test trajectories and heat maps showing locomotor activity patterns of mice in the indicated groups. Quantification of the time spent in the center zone (**C**) total distance traveled (**D**) and average movement speed (**E**) during the open-field test, $n = 7$. **F** Schematic illustration of the rotarod test used to assess motor coordination and balance. **G** Quantification of latency to fall in the rotarod test under the indicated conditions, $n = 7$. **H** Schematic illustration of the four-limb grip strength test. Quantification of average four-limb grip strength (**I**) and maximum grip strength (**J**) $n = 7$. **K** Representative immunoblot images showing the expression of apoptosis-related proteins in perihematomal brain tissue from mice subjected to ICH with or without FG-7142 treatment. β-actin serves as a loading control. **L–N** Quantification of protein expression levels shown in panel (**K**) $n = 3$. **O** Representative TUNEL staining images of brain sections from mice in the indicated groups, showing apoptotic cells in perihematomal regions. Nuclei are counterstained with Hoechst. Scale bar, 100 μm. **P** Quantification of TUNEL-positive cells shown in panel (**O**) expressed as the percentage of TUNEL-positive cells relative to total nuclei. Data are presented as mean ± SD. Statistical significance is indicated as described above.

Immunoprecipitation of TSPOAP1 successfully pulled down ANT2, whereas no interaction was detected between TSPOAP1 and CypD, a canonical CsA-sensitive regulator of the mPTP (Fig. 5E). Reciprocal Co-IP experiments further confirmed the association between TSPOAP1 and ANT2, supporting the presence of TSPOAP1 and ANT2 within the same protein complex. Immunofluorescence analysis further showed that endogenous TSPOAP1 displayed greater spatial overlap with ANT2 than with CypD, with Pearson correlation coefficients of 0.81 and 0.19, respectively (Fig. 5F). The FLAG-tagged TSPOAP1 C-terminal fragment also showed partial colocalization with ANT2, with Pearson correlation coefficients of 0.48, respectively, whereas its correlation with CypD was lower ($r = 0.37$) (Fig. 5G). Together, these imaging data support preferential spatial association of TSPOAP1 with ANT2 compared with CypD, while avoiding inference of precise submitochondrial localization.

Upregulation of TSPOAP1 by FG-7142 confers neuroprotection in vitro and in vivo

Given the protective role of TSPOAP1, we next sought to identify a compound capable of increasing TSPOAP1 expression. Molecular docking analysis using the full-length TSPOAP1 model predicted a potential interaction between FG-7142 with a best docking score of −6.80 kcal/mol (Fig. S4A–C). In addition, FG-7142 increased TSPOAP1 protein expression in WT cells (Fig. S4D, E). Based on these observations, we further examined whether FG-7142 could alleviate Hb-induced neuronal and mitochondrial injury and whether its protective effects were associated with the presence of TSPOAP1.

FG-7142 reduced the proportion of TUNEL-positive cells after Hb treatment (Fig. 6A, B). Consistently, immunoblot analysis showed decreased Bax and cleaved caspase-3 levels, along with partial preservation of Bcl-2 (Fig. 6C–F). Annexin V/PI flow cytometry demonstrated an increased viable cell fraction in FG-7142–treated WT cultures (Fig. 6G, H). JC-1 imaging further showed that FG-7142 attenuated Hb-induced mitochondrial depolarization, as indicated by preserved red fluorescence (Fig. 6I). Quantification of the JC-1 aggregate/monomer ratio confirmed significant restoration of $\Delta\Psi m$ in WT cells (Fig. 6J).

To assess the contribution of TSPOAP1 to these effects, we evaluated FG-7142 in Tspoap1$^{-/-}$ cells. In contrast to WT cells, FG-7142 did not significantly improve cell viability in knockout cultures following Hb exposure (Fig. 6K, L). JC-1 flow cytometry and confocal imaging similarly showed no significant restoration of $\Delta\Psi m$ in Tspoap1$^{-/-}$ cells (Fig. 6M, N). Together, these findings indicate that the protective effects of FG-7142 against Hb-induced mitochondrial and apoptotic injury require TSPOAP1. Moreover, SR48968, an reported TSPOAP1-upregulating candidate [44], increased TSPOAP1 expression and reduced Hb-induced Bax expression, supporting an inverse relationship between TSPOAP1 abundance and apoptotic vulnerability (Figure. S4F, G).

To evaluate the in vivo effects of FG-7142, mice were subjected to ICH and treated according to the experimental timeline (Fig. 7A). Behavioral performance was assessed using open field, rotarod, and grip strength tests. In the open field test, FG-7142–treated ICH mice showed increased time in the center zone, greater total distance traveled, and higher average speed compared with vehicle-treated ICH mice (Fig. 7B–E). Rotarod testing demonstrated prolonged latency to fall (Fig. 7F, G). Grip strength measurements showed significant improvement in both average and maximal four-limb strength (Fig. 7H–J). At the molecular level, perihematomal tissue analysis revealed reduced Bax and cleaved caspase-3 expression and increased Bcl-2 levels in FG-7142–treated mice (Fig. 7K–N). TUNEL staining further showed decreased neuronal apoptosis in the peri-lesional region (Fig. 7O, P). Collectively, these results indicate that FG-7142 improves neurological outcomes and reduces apoptosis following ICH in vivo.

Discussion

In this study, we identify TSPOAP1 as a previously unrecognized mitochondrial protective factor that constrains neuronal apoptosis. Despite its nomenclature and historical association with TSPO, TSPOAP1 appears to play a distinct role in ICH. TSPO exhibits context-dependent roles in microglial mitochondria: it supports Aβ clearance in Alzheimer's disease models but drives inflammatory injury through retrograde signaling in neuroinflammation [45, 46]. In our ICH

study, TSPOAP1 was predominantly detected in neurons and was associated with protection against mitochondria-mediated apoptosis. Nevertheless, contributions from microglia and other non-neuronal cells to the in vivo phenotype cannot be excluded. Historically, TSPO was incorporated into models of the mitochondrial permeability transition pore and was proposed to function as a structural component or direct regulator of the mPTP [47, 48]. However, more recent genetic, structural, and biochemical evidence no longer supports an obligatory role for TSPO in the mPTP formation or gating [49, 50], with recent studies instead implicating TSPO in distinct cholesterol- and porphyrin-related functions that are independent of direct mPTP regulation [51]. Interestingly, our present finding together with emerging evidences suggest that TSPOAP1 may be a regulator of mPTP and influence mitochondrial permeability transition [37, 38]. Notably, TSPOAP1 peaked at day 3, coinciding with increased Bax and disrupted Bcl-2/Bax balance [52], supporting its inverse temporal relationship with mitochondrial apoptotic vulnerability. Functionally, loss of TSPOAP1 exacerbated neuronal susceptibility to hemorrhagic stress, manifested by mitochondrial depolarization, enhanced mPTP opening, and subsequently engaged the intrinsic apoptotic machinery. Together, these observations identify TSPOAP1 down-regulation as an early and functionally relevant event in neuronal apoptosis following ICH.

We propose that TSPOAP1 influences mitochondrial apoptosis by modulating susceptibility to permeability transition. Although no canonical mitochondrial targeting sequence has been identified in full-length TSPOAP1 to date, our results indicate TSPOAP1 localize within the mitochondrial. Rather than participating in the conventional CypD-dependent gatekeeping machinery [53, 54], TSPOAP1 appears to regulate mPTP kinetics through a distinct molecular interface. Through a combination of in silico modeling and in vitro binding assays, ANT2 emerged as a critical interactor that physically associates with TSPOAP1. Notably, the functional and transcriptional coupling between TSPOAP1 and ANT2 suggests that this pair forms a specialized regulatory unit. This association is essential for maintaining mitochondrial membrane integrity, providing a mechanistic basis for how TSPOAP1 buffers neurons against hemorrhagic stress. Given accumulating evidence that ANT can modulate mPTP opening independently of CypD [55–58], our results position the TSPOAP1–ANT2 relationship as a potential ANT2-associated regulatory mechanism influencing mitochondrial fate under hemorrhagic stress.

Our findings indicate that enhanced mPTP susceptibility in TSPOAP1-deficient cells is accompanied by increased mitochondrial oxidative stress. While Hb stimulation elevated ROS generation in WT neurons, TSPOAP1-deficient cells exhibited increased mitochondrial H_2O_2 accumulation, which was further enhanced after Hb exposure, together with aggravated mPTP opening and mitochondrial depolarization. However, DCFH-DA fluorescence did not increase correspondingly, suggesting compartment-specific redox changes rather than a uniform rise in cellular oxidation. Whether mitochondrial ROS is a cause or consequence of mitochondrial dysfunction remains unresolved [29, 59–61]. Furthermore, the failure of CsA-mediated CypD blockade to alleviate mPTP dysregulation or apoptotic signaling in the knockout group aligns with emerging evidence that pore kinetics can be governed via CypD-independent routes, potentially involving ANT-dependent structural transitions under stress [62, 63]. Together, these findings support a non-classical mode of mPTP regulation associated with TSPOAP1 loss, characterized by increased mitochondrial oxidative stress and relative insensitivity to CypD inhibition.

In this study, we provide the first evidence that administration of FG-7142 confers neuroprotection in ICH model. Mechanistically, FG-7142 robustly upregulates TSPOAP1 expression, and the complete loss of its protective effects in TSPOAP1-deficient neurons establishes TSPOAP1 as an essential molecular mediator of FG-7142-induced neuroprotection. Previous studies have demonstrated that targeting mitochondrial dysfunction is an effective strategy for mitigating secondary neuronal injury following hemorrhagic or ischemic insults [64–66]. In this context, FG-7142 represents a small-molecule compound that reduces mitochondrial apoptosis during secondary brain injury. FG-7142 has been primarily investigated for its well-documented brain penetrance and its effects on neuronal excitability and survival in models of central nervous system stress and affective disorders [67–69]. Our findings extend these observations by establishing FG-7142 as a previously unrecognized modulator of mitochondrial dysfunction and neuronal apoptosis after ICH through TSPOAP1.

Several limitations of the present study warrant consideration. Although we demonstrate a physical and functional association between TSPOAP1 and ANT2, the precise structural determinants and molecular interfaces underlying this interaction remain undefined. Whether TSPOAP1 modulates mPTP permeability by directly altering ANT2 conformation, influencing higher-order protein complex assembly, or reshaping mitochondrial membrane

microdomains requires further investigation. In addition, although our data strongly support a CypD-independent mechanism of mPTP regulation downstream of TSPOAP1 loss, we cannot fully exclude potential contributions from other proposed pore components, such as F_0F_1-ATP synthase, under specific metabolic or pathological conditions.

In summary, our study identifies TSPOAP1 as a neuronal mitochondrial apoptosis regulator after ICH. TSPOAP1 loss was associated with increased mitochondrial oxidative stress, enhanced permeability-transition susceptibility, mitochondrial depolarization, and apoptotic signaling through an ANT2-associated mechanism. Biochemical, imaging, and computational analyses support an association between TSPOAP1 and ANT2. FG-7142 increased TSPOAP1 expression and attenuated mitochondrial and apoptotic injury, with these effects being markedly reduced in TSPOAP1-deficient neurons. Together, these findings identify TSPOAP1 as a potential regulator of mitochondrial resilience after ICH and support further investigation of the TSPOAP1–ANT2 pathway as a therapeutic target for secondary brain injury.

Materials and methods

Human samples

Perihematomal brain tissues were obtained from pediatric patients with intracerebral hemorrhage (ICH, $n = 3$). Control brain tissues were collected from age-matched children undergoing neurosurgical procedures for non-hemorrhagic conditions, using tissue unavoidably resected during surgical access ($n = 3$). The use of human tissue samples was approved by the Ethics Committee of the Children's Hospital of Chongqing Medical University (Approval No. 2021295). Written informed consent was obtained from the legal guardians of all pediatric participants before tissue collection.

Animals and ICH model

Adult male C57BL/6 mice (8 weeks old) were housed under standard laboratory conditions with ad libitum access to food and water. All animal procedures were approved by the Institutional Animal Care and Use Committee of Chongqing Medical University (IACUC-CQMU-2023-026). Experimental intracerebral hemorrhage (ICH) was induced by stereotactic injection of autologous blood (30 μL) into the right basal ganglia (AP − 0.2 mm, ML − 2.0 mm, DV − 3.5 mm) at a rate of 1.5 μL/min. Sham-operated mice received an equivalent volume of sterile saline.

Cell culture and treatments

HT22 mouse hippocampal neuronal cells were purchased from Solarbio Science & Technology Co., Ltd. Cells were maintained under standard culture conditions. Cultures were routinely treated with a mycoplasma removal and prevention reagent (Beyotime Biotechnology) according to the instructions. HT22 cells were cultured under standard conditions. Hemoglobin (Hb) was used to establish an in vitro hemorrhagic injury model. Cyclosporin A (CsA) was applied as a pharmacological modulator of mPTP opening. FG-7142 (MedChemExpress) was used as an intervention compound. Vehicle-treated cells served as controls. Detailed concentrations and treatment durations are provided in figure legends.

Construction and expression of FLAG-TSPOAP1-C

A mouse TSPOAP1 C-terminal fragment corresponding to amino acids 1561–1846 was designed and cloned into a FLAG-tagged expression vector to generate FLAG-TSPOAP1-C. The inserted sequence was verified by Sanger sequencing. HT22 cells were transfected with the construct using PEI (DNA: PEI = 1:4) according to instructions and

were collected 72 h after transfection for mitochondrial fractionation, proteinase K protection assays, and immunofluorescence analysis. The plasmid map and inserted sequence are provided in the Supplementary Materials.

Proteinase K protection assay

Mitochondria were isolated from N2A cells expressing a FLAG-tagged mouse TSPOAP1 C-terminal fragment encompassing amino acids 1561–1846 (FLAG-TSPOAP1-C). Equal amounts of isolated mitochondria were divided into untreated, proteinase K-treated, and proteinase K plus Triton X-100-treated groups. Mitochondria were incubated with proteinase K at the indicated concentration for 25 μg/mL 10 min at 37 °C. For membrane permeabilization, Triton X-100 was added to a final concentration of 1% before proteinase K digestion. Reactions were terminated by adding PMSF, followed by immunoblot analysis of FLAG-TSPOAP1-C, TOM20, and COX IV. TOM20 and COX IV were used as outer- and inner-mitochondrial membrane controls, respectively.

Hb injury model and FG-7142 and SR48968 treatment

For in vitro injury, cells were treated by Hb at a final concentration of 40 μM for 24 h, based on preliminary dose–response experiments. FG-7142 and SR48968 was dissolved in DMSO and diluted in culture medium immediately before use. Cells were pretreated with FG-7142 for 1 h prior to Hb exposure and maintained in its presence throughout stimulation.

Generation of TSPOAP1 knockout HT22 cells

TSPOAP1-knockout HT22 cells were generated using a CRISPR/Cas9 lentiviral system. Cells were transduced with sgRNA-expressing lentiviruses, selected with puromycin, and monoclonal populations were isolated. Gene disruption was confirmed by Sanger sequencing, immunoblotting, and quantitative PCR.

RNA sequencing and bioinformatic analysis

RNA sequencing was performed on human perihematomal tissues, mouse ICH and sham tissues, and WT or TSPOAP1-deficient HT22 cells with or without Hb stimulation by a commercial service provider. Differentially expressed genes were identified using standard pipelines with an adjusted P value (FDR) < 0.05. Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment analyses were conducted to identify significantly enriched biological processes and signaling pathways.

The raw mouse mRNA-sequencing data have been deposited in the National Center for Biotechnology Information Sequence Read Archive under BioProject accession number PRJNA1048733 and human perihematomal tissue transcriptomic data in GEO under GSE338415.

Assessment of apoptosis and cell viability

Cell viability and apoptosis were assessed using Calcein-AM/ propidium iodide (PI) staining, TUNEL assay, and Annexin V–FITC/PI flow cytometry according to the manufacturers' instructions. Quantitative analyses were performed using fluorescence microscopy, microplate reader measurements, or flow cytometry as appropriate.

Mitochondrial functional assays

$\Delta\Psi$m was assessed using the JC-1 assay by flow cytometry and confocal microscopy. mPTP opening was evaluated using the Calcein–$CoCl_2$ quenching assay, with fluorescence microscopy and flow cytometry used for quantification. Where indicated, ionomycin or carbonyl cyanide m-chlorophenyl hydrazone (CCCP) served as control treatments.

Measurement of ROS

Intracellular ROS were measured using DCFH-DA, and mitochondrial H_2O_2 levels were assessed using MitoPY1, Fluorescence intensity was quantified by flow cytometry and expressed as mean fluorescence intensity.

Measurement of ATP

Cellular ATP content was measured using a luciferase-based ATP assay kit according to the Beyotime's instructions. Luminescence was recorded using a microplate reader and normalized to total protein concentration determined by the BCA assay. Results were expressed as nmol/mg protein.

Mitochondrial fractionation and ultrastructural analysis

Mitochondrial and cytosolic fractions were isolated using a commercial mitochondria isolation kit. Fraction purity was assessed using VDAC1 and β-actin as mitochondrial and cytosolic markers, respectively. Mitochondrial ultrastructure was examined by transmission electron microscopy following standard fixation and embedding, and sectioning procedures.

Protein–protein interaction analyses

PPI networks were analyzed using the STRING database with the highest confidence score (0.900). Structural feasibility of TSPOAP1–ANT2 interaction was evaluated using GRAMM docking analysis, and docking models were visualized using PyMOL.

Molecular docking of FG-7142 with TSPOAP1

The three-dimensional structure of FG-7142 was obtained from PubChem and energy-minimized before docking. A full-length TSPOAP1 structural model was used as the receptor. Blind molecular docking was performed using GRAMM, and the top-ranked binding pose was selected according to the predicted binding energy. Protein–ligand interactions were visualized using PyMOL. The predicted binding site involved Arg920 and Thr923 within the second fibronectin type III domain, with a predicted binding energy of −6.80 kcal/mol.

Co-immunoprecipitation and immunofluorescence (IF)

Co-immunoprecipitation was performed using antibodies against TSPOAP1, Flag, CypD, or ANT2, followed by immunoblot analysis. Immunofluorescence staining were conducted using confocal microscopy. For colocalization analysis, endogenous TSPOAP1 or FLAG-TSPOAP1-C was co-stained with TOM20, ANT2, or CypD as indicated. Confocal images were acquired using identical laser power, gain, and exposure settings across comparison groups. Pearson's correlation coefficients were calculated from regions of interest using the Coloc 2 plugin in Fiji without intensity thresholding.

FG-7142 administration in vivo

FG-7142 was dissolved in DMSO and diluted in 20% sulfobutylether-β-cyclodextrin in saline. Mice received intraperitoneal injections of FG-7142 (10 mg/kg) once daily for three consecutive days after ICH induction. Vehicle-treated mice served as controls.

Behavioral assessments

A separate cohort of mice was used for behavioral experiments. Grip strength, rotarod performance, and open field activity were assessed at baseline and at 3 and 7 days after ICH by investigators blinded to treatment groups.

Experimental design, randomization, blinding, exclusion criteria, and sample-size determination

An initial sample size of 10 mice per group was planned based on our previous experience with the experimental ICH model, the anticipated requirements for behavioral testing, immunoblotting, and histological analyses, and the expected model-related mortality. Animals from the same experimental cohort were allocated to different outcome assessments according to the predefined experimental design. Therefore, the final number of animals used for each assay varied depending on the experimental endpoint, and the exact sample size for each analysis is provided in the corresponding figure legend.

Animals were randomly allocated to the experimental groups before surgery and pharmacological treatment. Behavioral assessments and quantitative image analyses were performed by investigators blinded to group allocation. For behavioral analyses, mice showing complete loss of spontaneous locomotion and an inability to perform the required behavioral tasks were excluded because the corresponding outcomes could not be technically assessed. All exclusions were documented before statistical analysis. No data points were excluded solely on the basis of the magnitude or direction of the experimental result.

Statistical analysis

All statistical analyses were performed using GraphPad Prism version 9.0. Comparisons between two independent groups were performed using an unpaired two-tailed Student's *t* test. When the assumption of equal variance was not met, Welch's *t* test was used. For non-normally distributed data, the Mann–Whitney U test was applied. Comparisons among three or more groups involving a single independent variable were performed using one-way ANOVA followed by Dunnett's multiple-comparisons test when multiple groups were compared with a common control, or Tukey's multiple-comparisons test when all pairwise comparisons were required. Experiments involving genotype and treatment were analyzed using two-way ANOVA followed by Šídák's multiple-comparisons test. Longitudinal behavioral data were analyzed using two-way repeated-measures ANOVA followed by Šídák's multiple-comparisons test. Data are presented as mean ± SD, with individual data points shown where applicable. All statistical tests were two-sided, and $P < 0.05$ was considered statistically significant.

Immunoblotting

For immunoblotting, cultured cells or perihematomal brain tissues were lysed in ice-cold RIPA lysis buffer supplemented with protease and phosphatase inhibitor cocktails. Lysates were incubated on ice and centrifuged at $12{,}000 \times g$ for 15 min at 4 °C to remove insoluble debris. Protein concentrations were determined using a bicinchoninic acid assay. Equal amounts of protein were separated by SDS–PAGE and transferred onto polyvinylidene difluoride membranes. Membranes were blocked with 5% non-fat milk in Tris-buffered saline containing 0.1% Tween-20 for 1 h at room temperature and then incubated with the indicated primary antibodies overnight at 4 °C. After washing, membranes were incubated with the appropriate horseradish peroxidase-conjugated secondary antibodies for 1 h at room temperature. Protein bands were visualized using an enhanced chemiluminescence detection system and quantified by densitometric analysis using ImageJ software. Target protein expression was normalized to the corresponding loading control. For mitochondrial and cytosolic fractions, VDAC1 or COX IV and β-actin were used as fraction-specific loading controls, as appropriate.

Detailed information on all primary and secondary antibodies used in this study, including vendors, catalog numbers, host species, applications, and working dilutions, is provided in Supplementary Table S5.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41419-026-09282-z.

Author contributions YD and CL conceived the study, designed the experiments, performed most of the in vitro and in vivo experiments, analyzed the data, and drafted the manuscript. YH and MZ contributed to animal model establishment, behavioral assessments, and histological analyses. YH assisted with molecular biology experiments, bioinformatic analyses, and data interpretation. XZ and HG supervised the study, provided conceptual guidance, acquired funding, and critically revised the manuscript. All authors reviewed and approved the final version of the manuscript.

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Data availability The data supporting the findings of this study are available from the corresponding authors upon reasonable request.

Competing interests The authors declare no competing interests.

Ethics The use of human brain tissue was approved by the Ethics Committee of the Children's Hospital of Chongqing Medical University (File No. 2021295) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from the legal guardians of all participants prior to tissue collection. All animal experiments were approved by the Institutional Animal Care and Use Committee of Chongqing Medical University (Approval No. IACUC-CQMU-2023-026) and were performed in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals. All efforts were made to minimize animal suffering and to reduce the number of animals used.

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